

EXPRESSION EVALUATION AND EXPRESSION CONVERSION

	Infix to Postfix	Infix to Prefix	Postfix to Infix	Prefix to Infix	Postfix to Prefix	Prefix to Postfix
Do we need to Reverse the given Equation?	NO	YES	NO	YES	NO	YES
Scan the given Equation	Left to Right	Left to Right	Left to Right	Left to Right	Left to Right	Left to Right
Initially Stack Should be Empty?	YES	YES	YES	YES	YES	YES
Rule of Associativity	Operators with same precedence cannot sit together in a stack (+, -, *, /, %, \$, ^ etc cannot possible) but In Case of ^ \$ ↑ Can sit together in a stack	Operators with same precedence can sit together in a stack (+, -, *, /, %, \$, ^ etc cannot possible) but In case of ^ \$ ↑ Cannot sit together in a stack	---	---	---	---
While Scanning if Operand comes	Direct Put that operand in OUTPUT STRING	Direct Put that operand in OUTPUT STRING	Push to the Stack	Push to the Stack	Push to the Stack	Push to the Stack
While Scanning if Operator comes	----	----	POP TWO recent elements from the TOP of the stack.	POP TWO recent elements from the TOP of the stack.	POP TWO recent elements from the TOP of the stack.	POP TWO recent elements from the TOP of the stack.
Sequence of POP	----	----	The 1st element which get pop is operand 2 (OP2) The 2nd element which get pop is operand 1 (OP1)	The 1st element which get pop is operand 1 (OP1) The 2nd element which get pop is operand 2 (OP2)	The 1st element which get pop is operand 2 (OP2) The 2nd element which get pop is operand 1 (OP1)	The 1st element which get pop is operand 1 (OP1) The 2nd element which get pop is operand 2 (OP2)
Operation to be performed		----	OP1 <Operator> OP2	OP1 <Operator> OP2	OP1 OP2 <Operator>	<Operator> OP1 OP2
Any Time What will be INSIDE THE STACK?	Operators Only (+, -, *, /, %, \$, ^ etc)	Operators Only (+, -, *, /, %, \$, ^ etc)	Operators And Operands Both	Operators And Operands Both	Operators And Operands Both	Operators And Operands Both
Parenthesis () required? In Final Output	NO	NO	YES	YES	NO	NO